hype check

Challenging Causal Foundation Models for Uplift

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BIO

Vadim Porvatov is a researcher working at the intersection of Causal AI and applied machine learning. He holds an MSc in Data Science from the University of Amsterdam.

His current focus is building practical tools that help financial teams move from predictive analytics to effect-based decision-making: **uplift modeling**, **causal discovery**, **A/B tests**, and **LLM-assisted analysis of causal insights**.

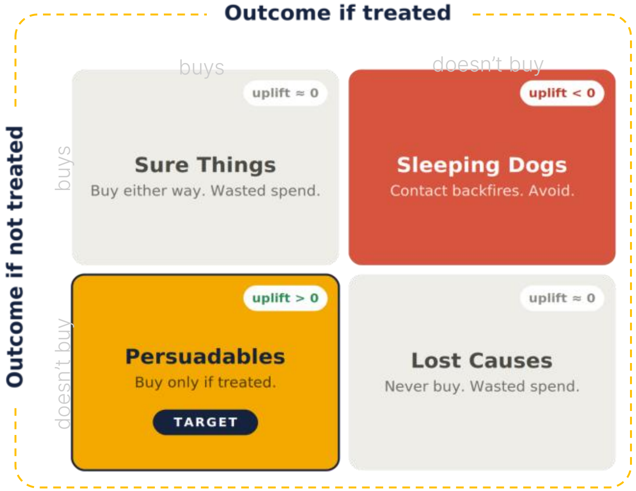
Before this, Vadim worked on **graph ML**, **recommender systems**, **computer vision** and **time-series models**.

RESEARCH SCOPE & PROBLEM STATEMENT

industrial example of uplift modeling task

our goal is to differentiate between users with high effect from marketing campaign (generating profit for a company) and others:

uplift(x) = P(Buy | Treated) - P(Buy | Not Treated)



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Project Scope

Uplift is a well-known ranking and policy decision task based on **effect estimation** (in comparison to the classic ML which **just predicts single outcome**). We investigate whether a ready-to-use causal foundation model (CausalPFN: paper, github) helps real targeting in three steps:

Check: untuned causal foundation models vs strong tuned baselines on the ranking metrics;
Break: map the regimes where foundation models advantages disappear;
Fix: develop a **CausalPFN-Rank**, a training-free adaptation of the studied foundation models (**smart context** + **output re-ranking**).

Why this is practically interesting?

1. In business context uplift is judged by ranking (e.g., Qini, uplift@k, AUUC), but CausalPFN is tuned for effect error metrics -- we expect that the best estimator is not the best ranking model.
2. Practical insights regarding when an untuned foundation model is enough vs the tuned one / specialized method / untuned + cheap fix.

HYPOTHESES

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We are going to test five hypotheses:

H1 Untuned, CausalPFN matches tuned non-foundation learners on semi-synthetic accuracy (PEHE) but not on real targeting (Qini)

H2 Ranking by accuracy correlates weakly with ranking by targeting value; the best estimator is often not the best targeter

H3 The advantage breaks in three regimes: data beyond the context window, a small control group, and low conversion

H4 Engineered context (treatment-balanced selection plus multi-context averaging) recovers much of the lost targeting value, with no training

H5 Re-ranking the outputs improves targeting even when accuracy is unchanged, narrowing the gap to specialist methods

GOALS & PLAN

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Goals

G1: reproducible uplift results over the dataset pool; reproduce CausalPFN's Hillstrom example as a smoke test;

G2: map the break-axes by controlled subsampling, with bootstrap CIs and the accuracy-vs-targeting correlation;

G3: build and ablate CausalPFN-Rank; report per-regime recovery and compute cost.

Plan (4 weeks)

- **W1**: harness, smoke test, baselines, first untuned comparison;
- **W2**: break-axis mapping; add Q-Learner and GP-CATE;
- **W3**: implement and ablate CausalPFN-Rank;
- **W4**: final runs with CIs and efficiency; write-up.

APPROACH & METHOD

CausalPFN-Rank (frozen model, no training)

- **Input**: treatment-balanced context and averaging over several retrieved contexts, vs the model's own default retrieval;
- **Output**: re-rank the scores, a calibrator trained for ordering, a cautious lower-bound score, and a ratio score for low conversion.

Baselines, data, metrics

- **Baselines**: S/T/X/DR on boosting, Causal Forest, BCF, DragonNet/DESCN; Q-Learner and GP-CATE on their regimes;
- **Data***: Criteo / X5 and Hillstrom (real RCT); IHDP / ACIC (semi-synthetic);
- **Metrics**: Qini/AUUC and uplift@k with bootstrap CIs; PEHE where ground truth exists; efficiency reported separately.

*datasets will be provided in a unified preprocessed format by the curator

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Thx!